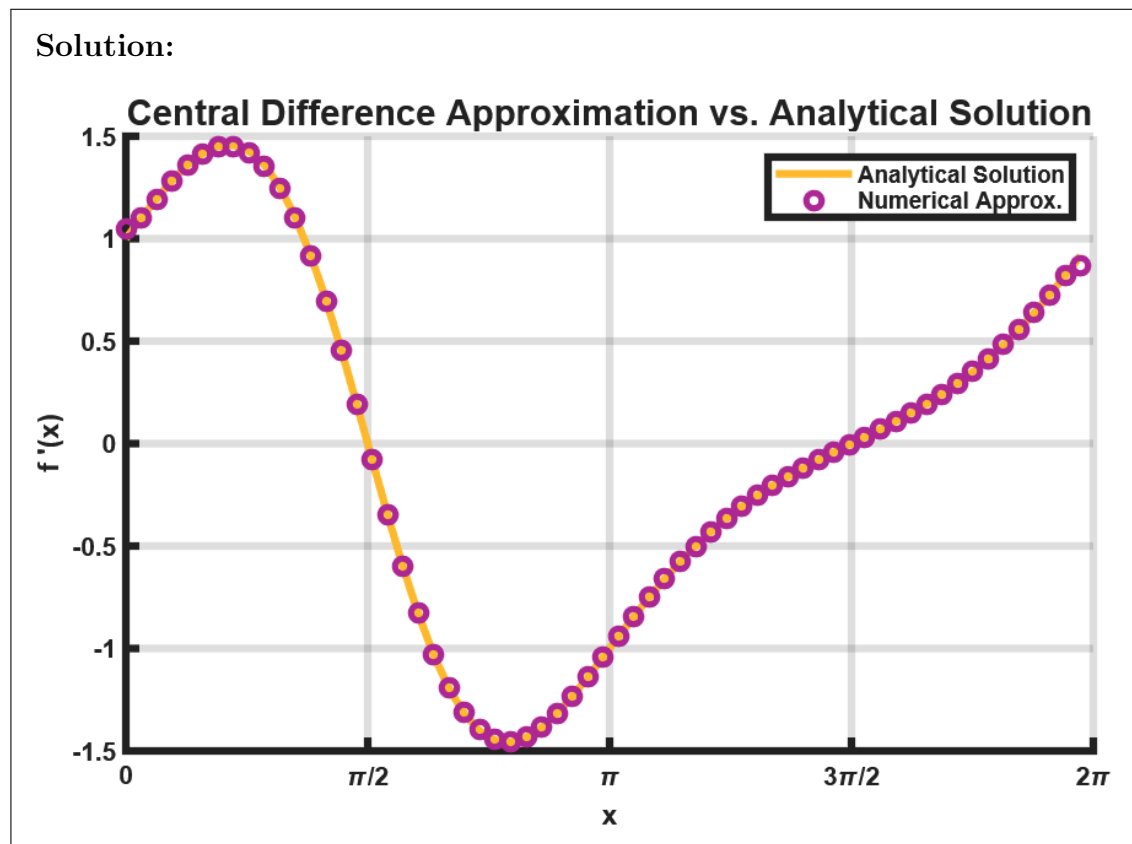


1. Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the interval $0 \leq x \leq 2\pi$. Use second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

- (a) Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.



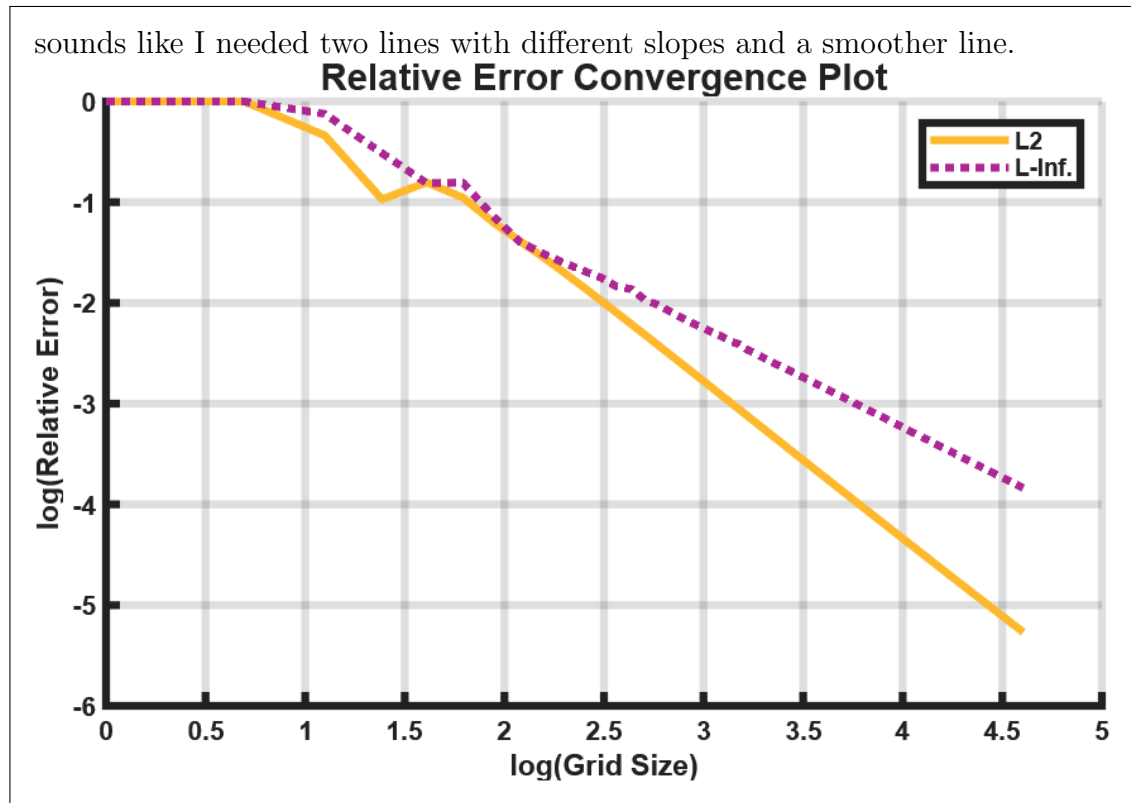
- (b) Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Measure relative error as

$$Err = \frac{\|f'(x) - f'_n(x)\|_p}{\|f'(x)\|_p}.$$

Plot the error using both the $p = \infty, 2$ norms in a log-log scale. The relative errors should be lines with slopes 1 and $3/2$ respectively, i.e. it scales with $\frac{1}{N}$, $\frac{1}{N^{3/2}}$. Extra Credit: The $3/2$ slope is strange, why is that happening?

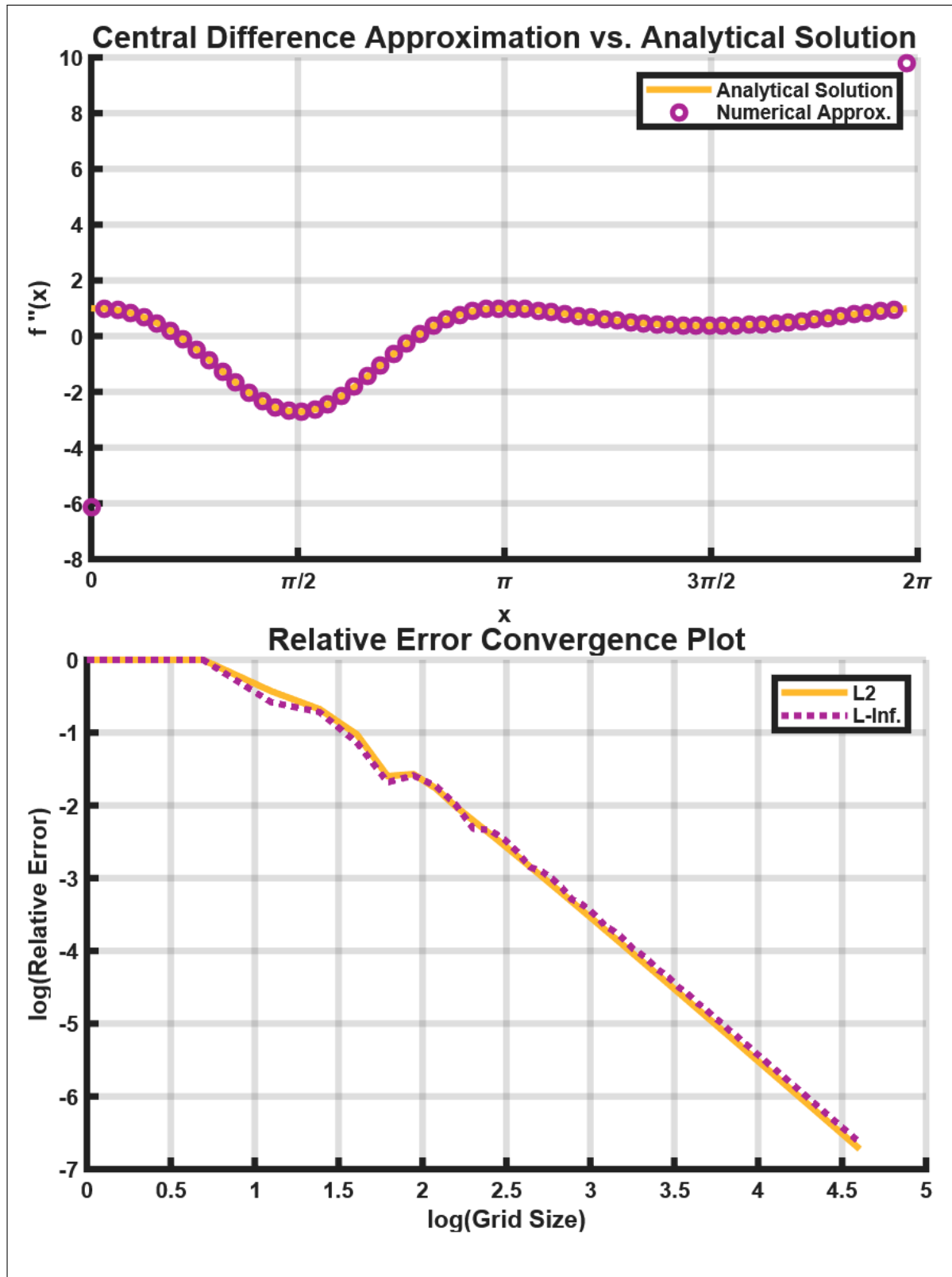
Solution:

As a general note, I was unsure of what this problem was asking me to plot. It



- Repeat problem 1 but compute the second derivative $f''(x)$. Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Solution:



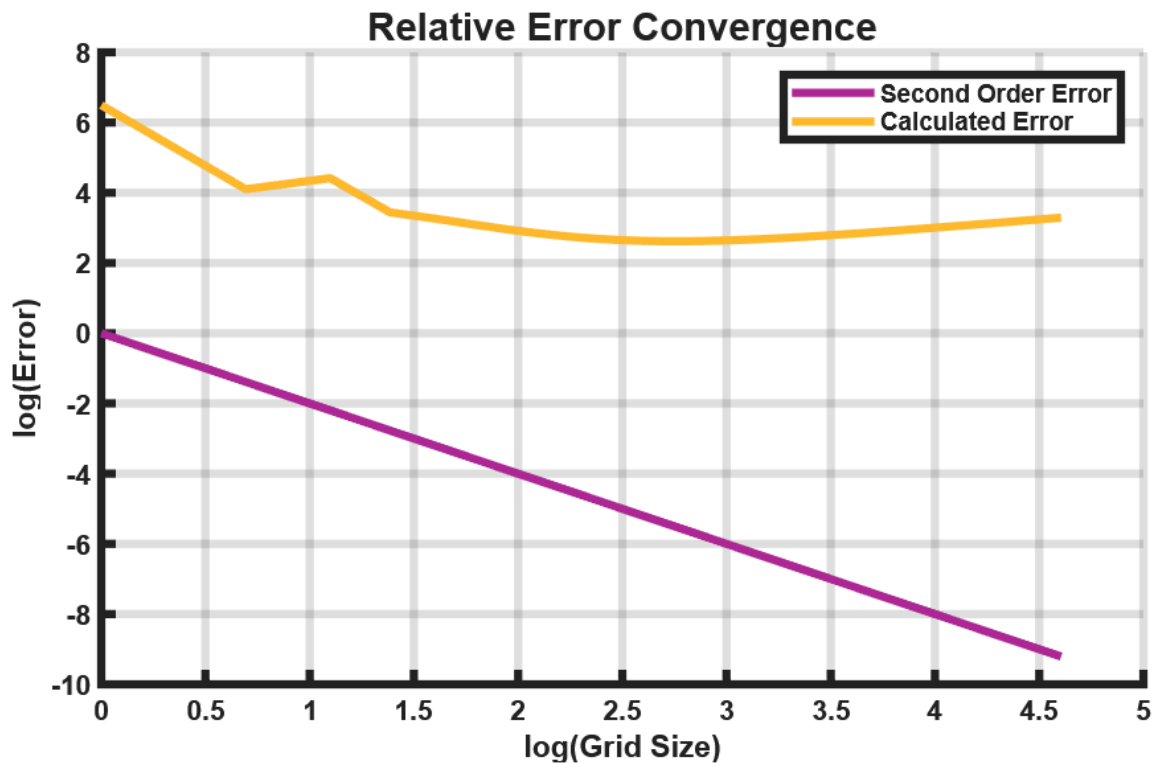
3. Consider the ODE

$$\frac{d^2u}{dx^2} + \sin(x) \frac{du}{dx} + u(x) = f(x)$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = \sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Solution:

Something's not right here but it's close.



4. Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5], y \in [0, 5]$. Use the test solution $u(x, y) = \sin(x) \cos(y)$.
- (a) Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).

Solution:

I have more of a start on this, but no plots generated yet.

-
- (b) Modify your code to apply Neumann conditions on all boundaries. This won't work. Why not? - both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Solution:

This would have been completed after part a, I think it's too late to finish this meaningfully and I should focus on the next homework.